

Phase-matching and mitigation of four-wave mixing in fibers with positive gain

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Abstract: We present a theoretical study of four-wave mixing interactions in fibers in the presence of gain. In contrast to passive fibers, positive gain at the pump wavelength leads to constructive generation of the signal and idler waves, even in the case of large phase-mismatch, so that FWM processes can be very efficient even in isotropic single-mode fibers with normal dispersion. We also propose simple ways to mitigate these parametric interactions by applying a controlled variation of the phase-mismatch along the fiber. These concepts apply to all optical amplifiers.

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1. Introduction

Pulsed fiber amplifiers have gained considerable interest because they are able to deliver nanosecond pulses with very high average and peak powers and diffraction-limited beam quality in highly reliable devices, which are major advantages for practical applications such as material processing and LIDAR. Recent advances in double-clad, Yb-doped, large-mode-area (LMA) fibers has led to a record combination of average and peak output powers at 1064nm [1, 2]. Different techniques have been implemented to ensure single-transverse-mode output while increasing the effective mode area of the fiber [3-6]. These approaches decrease

the peak irradiance and the fiber length, thereby raising the threshold for nonlinear process and increasing the attainable output power and pulse energy.

Despite these advances, nonlinear processes remain the limiting factor for power scaling because of practical limitations on the core diameter and the very high peak powers generated in the fiber (>1 MW, corresponding to a peak irradiance of up to 440 GW/cm^2). The limiting nonlinear processes depend on the pulse duration and spectrum. For relatively long pulses (>2 ns) with a transform-limited linewidth, stimulated Brillouin scattering (SBS) is dominant. For pulses of ~ 1 ns duration, stimulated Raman scattering (SRS) dominates. For pulse durations below ~ 0.5 ns, self-phase modulation induces large distortions in the spectrum. Recent results indicate that a pulse duration of 1 ns is suitable for maximizing the pulse energy while minimizing these nonlinear effects. In that case, four-wave mixing (FWM) is observed, which converts two photons of the amplified beam into two new frequencies [7], leading to emission over a relatively broad spectrum [8]. This effect is efficient only if the four interacting fields remain phase-matched as they propagate in the fiber. Phase-matching can occur via many different mechanisms, e.g., modal dispersion in multimode fibers, use of a pump wavelength with anomalous dispersion or close to the zero-dispersion wavelength, or use of birefringent fibers [7]. This is for example the case of fiber lasers and amplifiers operating in the $1.5\mu\text{m}$ region, where FWM limits the peak power. However, in single-mode isotropic fibers with normal dispersion, which applies to most pulsed fiber amplifiers operating around $1\mu\text{m}$ wavelength, phase-matching cannot be achieved; FWM thus remains very weak and was not previously considered as a limiting process, even though it has been observed [8].

In this paper, we present what is to our knowledge the first study of the phase-matching properties of FWM in fibers in the presence of gain. We point out the existence of a unique and previously unrecognized phenomenon, “gain-induced phase-matching”, which allows efficient FWM even for an interaction that is nominally highly phase-mismatched in isotropic single-mode fibers with normal dispersion. We show that FWM can become the limiting factor in the nanosecond temporal regime. The solution of the wave equation allows us to propose simple approaches to efficiently mitigating this detrimental nonlinear process by applying a controlled longitudinal variation of the phase-mismatch along the fiber.

2. FWM in nanosecond amplifiers

2.1. Nonlinear equations

We focus on partially degenerate FWM in isotropic fibers: $\omega_1 + \omega_1 = \omega_3 + \omega_4$ (ω_1 is the amplified beam, called the pump, and ω_3 and ω_4 are the generated signal and idler waves, $\omega_3 < \omega_4$). For concreteness, we consider the following amplifier parameters, which are typical for modern LMA fibers employed in a master oscillator – power amplifier configuration: silica fiber with an effective core area $A_{\text{eff}} = 500 \mu\text{m}^2$ (core diameter $\sim 30 \mu\text{m}$), $n_2 = 2.7 \cdot 10^{-20} \text{ m}^2/\text{W}$, power gain coefficient of $g = 0.5 \text{ m}^{-1}$ (20 dB over 9.2 m, assuming simple exponential gain), seed pulses at 1064 nm of $5 \mu\text{J}$ and 1 ns duration (FWHM). Estimation of the respective characteristic lengths allowed us to determine the relevant effects in the regime of nanosecond amplifiers. With the above parameters and the definitions of [7], we neglect dispersion, walk-off, and time-dependent phase shifts; this simplifies the equations by eliminating all time-derivatives. Furthermore, it was shown in previous experiments that SBS is insignificant for pulses of ≤ 1 ns duration [2,8] due to pulse walk-off and the large signal bandwidth. For 1 ns pulses, SRS can be described as quasi-cw; using the formalism of Ref. [7], the equations reduce to:

$$\begin{cases} \frac{\partial E_1}{\partial z} = j\gamma \left[|E_1|^2 + (2-\rho) \left(|E_3|^2 + |E_4|^2 \right) \right] E_1 - \frac{g_R}{2A_{\text{eff}}} |E_3|^2 E_1 + \frac{g}{2} E_1 + j\gamma E_1^* E_3 E_4 e^{j\Delta k z} \\ \frac{\partial E_3}{\partial z} = j\gamma \left[|E_3|^2 + (2-\rho) \left(|E_1|^2 + |E_4|^2 \right) \right] E_3 + \frac{g_R}{2A_{\text{eff}}} |E_1|^2 E_3 + j\gamma E_4^* E_1 E_4 e^{-j\Delta k z} \\ \frac{\partial E_4}{\partial z} = j\gamma \left[|E_4|^2 + (2-\rho) \left(|E_1|^2 + |E_3|^2 \right) \right] E_4 + j\gamma E_3^* E_1 E_4 e^{-j\Delta k z} \end{cases} \quad (1)$$

with the complex electric field $E_i(z)=A_i(z)e^{j\phi(z)}$ related to power by $P_i(z)=A_i^2(z)$. The terms in the right-hand side of the first equation describe respectively self- and cross-phase-modulation, SRS, amplification gain, and FWM. g_R is the Raman gain and ρ is the fractional Raman contribution (typically $\rho=0.18$ in silica), the nonlinear coefficient is $\gamma=n_2\omega_1/cA_{\text{eff}}$. Δk is the phase-mismatch. We assume single-mode propagation and so neglect the waveguide contribution [9], so $\Delta k=2\pi(n_3/\lambda_3+n_4/\lambda_4-2n_1/\lambda_1)$. We neglect Raman contribution to FWM.

Assuming non-depletion of the pump field during FWM ($E_{3,4}<E_1 \forall z$), the first equation is solved to yield $E_1(z)=\sqrt{P_0} \exp(gz/2)\exp(j\gamma P_0 e^{gz}/g)$ with $P_0=|E_1(0)|^2$ the seed peak power. Equations for $E_{3,4}$ are transformed by setting $B_3=E_3 \exp(-j2\gamma P_0 e^{gz}/g)$. The net phase-mismatch and the signal field obey:

$$\kappa(z)=\Delta kz+2\frac{\gamma P_0}{g}e^{gz} \quad (2)$$

$$\frac{d^2 B_3}{dz^2} + [j(\Delta k + 2\gamma P_0 e^{gz}) - g] \frac{dB_3}{dz} - \gamma^2 P_0^2 e^{2gz} B_3 = 0 \quad (3)$$

In the limit of passive fibers, $g=0$, the above relations are similar to equations of Ref. [7]. The signal field is efficiently generated along the fiber only if $0 > \Delta k > -4\gamma P_0$; in the case of normal dispersion, the net phase-mismatch exhibits a linear variation along the propagation, and FWM is not efficient whatever the value of the pump field P_0 .

A major consequence of positive gain g is that the coherence length of FWM varies along the fiber ($L_c(z) \approx 2\pi/[\Delta k + 2\gamma P_0 e^{gz}]$). In order to derive analytical solutions for $E_{3,4}(z)$, we note that the variation of the pump field along the fiber is much slower than the oscillation of the fields due to non-phase-matched FWM (i.e. since $g \ll \Delta k + 2\gamma P_0 e^{gz}$, for typical values of the parameters the characteristic length for amplification $L_g \sim 1/g \sim 1\text{m}$ is much longer than the coherence length $L_c \sim 1\text{cm}$), so we approximate $P_0 e^{gz}$ to a constant. This approximation is validated *a posteriori* by the agreement between analytical relations with the exact numerical solutions of Eq. (1). The generated signal power deduced from Eq. (3) is then of the form:

$$P_3(z) = [A \exp(u_+ z) + B \exp(u_- z)]^2 \quad (4)$$

$u_{\pm}$ take simple forms in the limits of large and small Δk , detailed in the next paragraph. Constants A and B depend on the input powers at fiber entrance. Because amplifiers are only seeded at the pump wavelength, the input signal and idler fields result from noise photons.

2.2. Asymptotic solutions, analysis of the generated fields

In the limits of large and small mismatches:

- $\Delta k \gg 2\gamma P_0 e^{gz}$. In this case, $g \ll \Delta k + 2\gamma P_0 e^{gz} \forall z$, $u_{\pm} \approx -j\Delta k/2 \pm \sqrt{[\gamma P_0 \exp(gz)]^2 - (\Delta k/2)^2}$ $u_{\pm}$ is purely imaginary, and in the limit of very large mismatch Eq. (4) becomes:

$$P_3(z) \propto \sin^2(\Delta kz/2) \quad (5)$$

which is the usual evolution of the signal in the case of non-phase-matched interaction in media without gain. The signal power oscillates (the period is constant in that case, $L_c \sim 2\pi/\Delta k$) and is not efficiently constructed along the propagation.

- $\Delta k \ll 2\gamma P_0 e^{gz}$. In this case, $u_{\pm} \approx g/2 - j\gamma P_0 e^{gz} \pm \sqrt{(\gamma P_0 e^{gz})^2 + (j2\gamma P_0 e^{gz} - g)^2}/4$; for

typical parameters, $g \ll 2\gamma P_0 e^{gz}$ for all z , and Eq. (4) is approximated by:

$$P_3(z) \propto \exp(gz) \quad (6)$$

This case is unique to the fiber with positive gain: the net mismatch (2) and the coherence length ($L_c(z) \sim \pi/(\gamma P_0 e^{gz})$ in this limit) vary nonlinearly with distance. Thus, the power flow from the pump to the signal and idler accumulated over the first part of one coherence length is no longer equal to the power flowing in the reverse way in the second part. As a consequence, the signal field never goes back to its initial value, but is progressively constructed as it propagates along the fiber, even with non-zero phase-mismatch in single-mode isotropic fibers with normal dispersion. It should also be noted that Raman amplification (not taken into account in the above expression) will enhance this effect for signal wavelengths close to the Stokes peak.

The above analytical expressions provide useful understanding of the important parameters; however they were derived through important assumptions and only apply in limit cases. In order to illustrate more accurately the effect of “gain-induced phase-matching,” Fig. 1 shows the evolution of the signal power at $\lambda_3=1094\text{nm}$ along the fiber, calculated from the numerical integration of Eq. (1) with the above parameters (input signal and idler are one photon each, $\Delta k=41.5\text{ m}^{-1}$). It demonstrates the critical impact of gain on the generation of the signal: the field at λ_3 is progressively constructed along the fiber, and the signal power reaches the same order of magnitude as the amplified pump power (dashed line) after a few meters. From this exact calculation, the variation of $P_3(z)$ is more complicated than analytical Eq. (4)-Eq. (6). Close to the fiber input, $\Delta k > 2\gamma P_0 e^{gz}$, so the curve has a pronounced oscillatory behaviour as in Eq. (5). The end of the fiber corresponds to the limit $\Delta k < 2\gamma P_0 e^{gz}$ so that the behaviour is close to an exponential shape, as expected from Eq. (6). Maximum amplification of the pump (12.2dB) is reached at $z=5.65\text{ m}$. Beyond that position, the undepleted-pump approximation does not hold. For comparison, the case of the same fiber with no gain ($g=0$, $g_R=0$) is also shown on the same graph; in that case, the signal field exhibits the usual sinusoidal oscillation, as expected from Eq. (5). Figure 1(b) shows more clearly the departure of the signal power from purely oscillatory behaviour: constructive generation of signal occurs from the entrance of the fiber as soon as the gain is non-zero, even with weak pump input. In contrast, in a passive fiber ($g=0$) the signal field at λ_3 keeps an oscillatory behavior even if the input pump power P_0 is 10 times larger. After 3m in the active fiber ($g=0.5\text{m}^{-1}$), the signal is 6.3 times larger while the pump at λ_1 is lower (58% compared to the passive fiber with stronger input). From these curves, we conclude that gain is required to achieve “phase-matching” in the case of normal dispersion, while large pump power alone is not sufficient [7].

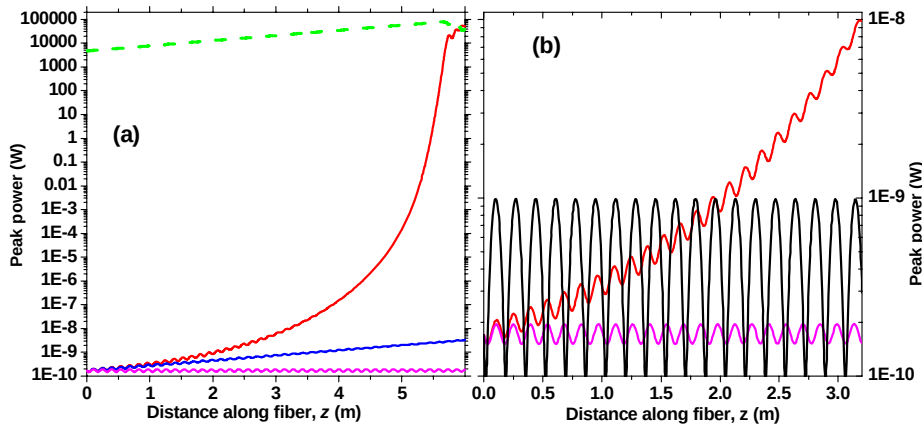


Fig. 1. Peak powers along propagation in the fiber: (a) Fixed seed energy; (b) Zoom towards fiber entrance. Solid curves are signal ($\lambda_3=1094\text{nm}$), green dashed curve is pump ($\lambda_1=1064\text{nm}$). Different values of seed energy, gain and Raman coefficient: red $\{E_{in}=5\mu\text{J}, g=0.5\text{m}^{-1}, g_R=5.10^{-14}\text{m}^2/\text{W}\}$; magenta $\{E_{in}=5\mu\text{J}, g=0\text{m}^{-1}, g_R=0\text{m}^2/\text{W}\}$; blue $\{E_{in}=5\mu\text{J}, g=0\text{m}^{-1}, g_R=5.10^{-14}\text{m}^2/\text{W}\}$; black $\{E_{in}=50\mu\text{J}, g=0\text{m}^{-1}, g_R=0\text{m}^2/\text{W}\}$.

Calculations at other signal wavelengths show similar qualitative behavior. For input signal and idler fixed at one noise photon each, the spectral dependence of FWM arises from the relative importance of Raman gain g_R and the phase-mismatch Δk ; the signal generated along the fiber varies as in Fig. 1, but depletion occurs at different fiber length or pulse energy (peak power). A detailed study of these effects is beyond the scope of this paper; experimental data and calculated spectra are reported in Ref. [10].

From the above analysis, a characteristic length for “gain-induced phase-matching” is:

$$L_{PM} = \frac{1}{g} \ln \left(\frac{\Delta k}{2\gamma P_0} \right) = \frac{1}{g} \ln \left(\frac{\Delta k \lambda_1 A_{eff}}{4\pi n_2 P_0} \right) \quad (7)$$

If the fiber length (L) is $\ll L_{PM}$, gain has no noticeable impact on phase-matching. In contrast, if $L \geq L_{PM}$, gain has a major effect on FWM. Using the parameters listed above and $\Delta k = 100 \text{ m}^{-1}$, ($\lambda_1 = 1064 \text{ nm}$ and $\lambda_3 \sim 1100 \text{ nm}$) then $L_{PM} = 6.9 \text{ m}$ and the corresponding gain at λ_1 is only 15dB. This example shows that FWM cannot be neglected in typical fiber amplifiers.

3. Mitigation of “gain-induced phase-matching” of FWM

From Eq. (7), “gain-induced phase-matching” remains negligible only if the output amplified power obeys $P_{out} \ll \Delta k \lambda_1 A_{eff} / 4\pi n_2$, which limits the power that can be extracted from the amplifier without distortions by the FWM process. As expected, this maximum power increases with the effective mode area (and hence the fiber core diameter) and with decreasing nonlinearity. Less intuitive, the maximum power scales with the phase-mismatch. Careful design of the fiber effective area and dispersion might then help mitigating FWM. However, current fiber-fabrication processes, practical handling considerations and non-availability of fused-fiber pump combiners with matched core diameters set limits for implementation in reliable all-fiber systems.

One approach to mitigating “gain-induced phase-matching” in standard LMA fibers is to introduce a longitudinal variation in the phase-mismatch, in the form $\Delta k_{eff}(z) = \Delta k + f(z)$. An analytical solution of Eq. (1) is obtained in the form of Eq. (4) in the limit of slowly varying $f(z)$. If $f(z)$ verifies $\Delta k + f(z) \gg 2\gamma P_0 e^{gz} \forall z \in [0, L]$, the signal power is given by a relation similar to Eq. (5), and is then not efficiently generated.

In order to validate this concept, Fig. 2 shows the signal power deduced from numerical integration of Eq. (1) in fiber with controlled longitudinal variation of the phase-mismatch and different functions $f(z)$.

This approach efficiently reduces the generated signal: in particular, the simplest case of a linear gradient of the phase-mismatch decreases the generated signal by up to 5 orders of magnitude, which has a major impact on the ability to extract more amplified power from the fiber. No depletion of the pump is observed for the considered fiber length, and the maximum amplification at λ_1 is 14.8dB at $z = 6.85 \text{ m}$; in this case, the limiting nonlinear process is no longer FWM but SRS, because the same maximum is found with $n_2 = 0$. Linear variation of Δk is thus sufficient to effectively eliminate the impact of FWM.

There are different physical means to implement this longitudinal variation of the net phase-mismatch, by altering the material dispersion, the modal dispersion, or even the nonlinear phase-mismatch. One can apply a thermal gradient along the fiber or optimize that due to absorption of the diode pump power in the amplifier. A longitudinally variable stress can be used: for coiled fibers, the bend radius can be varied along the fiber, either continuously or discontinuously (including periodically). Varying the bend radius also influences modal propagation constants and hence Δk . One equivalent is a fiber with a helical core [6] but non-uniform helical trajectory. Another alternative is to apply periodic changes, for example by alternating portions of fibers with different dispersions, or leaving uncoiled sections of the fiber between coiled ones [11]. A longitudinal taper of the fiber core diameter

is another way for controlling the modal dispersion. Exposure of the fiber to ultraviolet radiation can vary the index and hence the modal propagation constants. The use of birefringent fibers might also be a convenient way to control the linear phase-mismatch Δk , and so to reduce the generation of the signal wave. Finally, a longitudinal variation of the nonlinear coefficient, $\gamma(z)$, or of the gain, $g(z)$, also varies the nonlinear phase-mismatch, $2\gamma P_0 e^{g z}$, and can mitigate FWM.

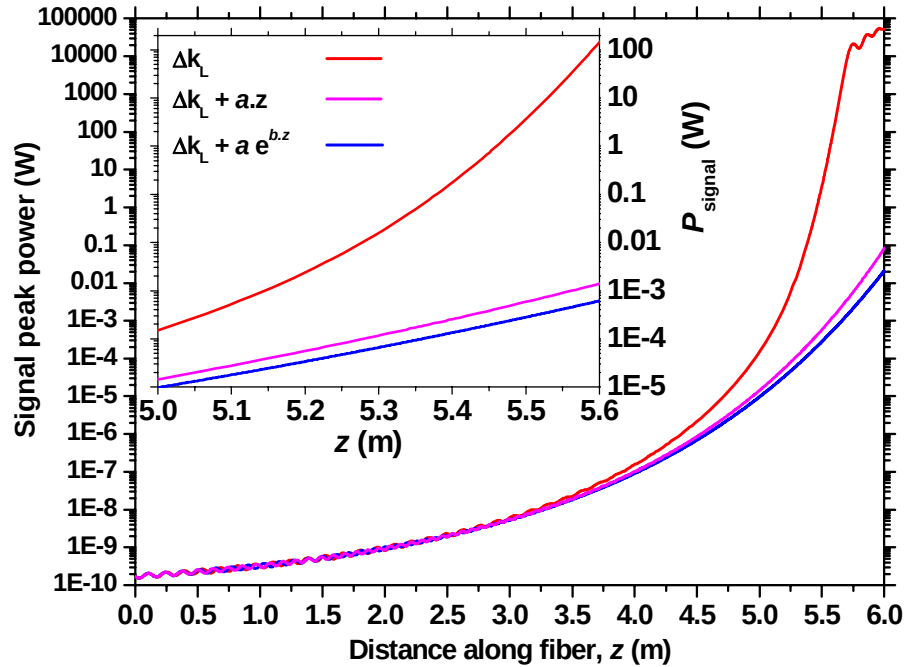


Fig. 2. Signal power along the fiber with longitudinally controlled phase-mismatch. Red: constant $\Delta k=41.5\text{m}^{-1}$. Magenta: linear increase of Δk , slope $a=4\pi\text{m}^{-1}$. Blue: exponential increase of Δk , $a=b=1.4\text{m}^{-1}$. Inset: zoom towards fiber exit.

4. Conclusion

We demonstrate, for the first time to our knowledge, the very unique behaviour of four-wave mixing in fiber amplifiers: positive gain at the pump wavelength leads to constructive generation of the signal and idler waves, even in the case of large phase-mismatch. We conclude that FWM processes can be very efficient even in isotropic single-mode fibers with normal dispersion. Practical examples show that this effect can be the most limiting one in pulsed fiber amplifiers operating in the nanosecond range. A detailed experimental study of these limitations will be reported in a separate paper.

Based on our theoretical analysis, we also propose simple ways to efficiently mitigate FWM by applying a controlled variation of the phase-mismatch along the fiber.

Finally, it is important to note that the equations are not specific to fibers, so that these concepts are generic to all optical amplifiers, including waveguides or bulk, or to fiber lasers.

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